

Call for Papers

Advances in Observation, Modeling, and Prediction of the Hydrosphere

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Accelerated climate change has introduced unprecedented instability into Earth's hydrological systems. To protect global water security, geoscientific research must shift from static models to integrated, predictive frameworks. This Collection highlights cutting-edge methods combining data-driven discovery and physical process modeling. We prioritize studies resolving multiscale hydrological processes—spanning local sub-daily variations to global hydroclimatic oscillations—to advance our predictive understanding of the hydrosphere.

This Collection aims to explore how next-generation Earth observation and computational frameworks can seamlessly integrate data-driven artificial intelligence with physically based hydrological models to forecast shifts in the hydrosphere under climatic uncertainty. It covers topics such as atmospheric water vapor circulation, groundwater-surface water interactions, terrestrial hydrological cycles, biosphere-hydrology couplings, and hydrosphere-lithosphere interactions mediated by geomorphic land-forming processes.

We welcome Original Research articles, Reviews, and Perspectives that address advanced observational and modeling strategies, as well as forecasting skills. Topics of interest include, but are not limited to:

- Next-Generation Observation and Artificial Intelligence for Hydrospheric Research
- Big Data Applications in Integrated Water Science
- Multi-Scale Changes in the Hydrosphere
- Drivers of Hydrospheric Changes and Human Intervention
- Couplings between Biospheric Processes and Hydrological Cycles
- Soil Moisture-Atmosphere Interactions
- Hydrosphere-Lithosphere Interactions
- Regional and Global Water Modeling
- Water System Monitoring and Forecasting
- Hydro-Induced Gravity Variations and Ground Deformations
- Freshwater Ecosystems under Global Change

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